# Curriculum Vitae - Li Hongmin

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**Primary Affiliation:** Sato Laboratory for Biomedical Data Science, Institute of Science Tokyo (Laboratory)

**Guest Affiliation:** Frith Lab, The University of Tokyo (Laboratory)

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## Research Interests

- AI-Automated Scientific Workflows
- AI for Science and Scientific Discovery
- AI Research and Evaluation
- Bioinformatics and Biological Sequence Analysis
- Biomolecular Sequence Design and Optimization

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## Education

- **2019.04 —2022.03** | **Ph.D. in Computer Science (Information Systems Engineering)** —University of Tsukuba
- **2017.04 —2019.03** | **Master's in Computer Science (Information Systems Engineering)** —University of Tsukuba
- **2011.09 —2015.07** | **Bachelor's in Electronic Information Engineering** —Ningxia University

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## Academic & industry appointments

- **2026.4 —Present** | **Researcher** —Sato Laboratory for Biomedical Data Science, Institute of Science Tokyo; Laboratory
- **2026.4 —Present** | **Guest Researcher** —Frith Lab, The University of Tokyo; Laboratory
- **2026.6 —Present** | **Founder** —AlphaScience Lab (Research community); Website
- **2023.5 —2026.3** | **Postdoctoral Researcher** —The University of Tokyo
- **2022.10 —2023.5** | **Machine Learning Engineer** —HAOMO.AI —Autonomous-driving project · 蓝色空间领航者 (A super high salary project)（类似华为天才少年计划）
- **2022.4 —2022.10** | **Postdoctoral Researcher** —The University of Tokyo; Details
- **2021.7 —2021.9** | **Machine Learning Engineer (Internship)** —HAOMO.AI

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## Research funding & computing support

- **2026.06** | **Google Cloud TPU Builders Award** —Cloud computing support · USD 7,500 in Google Cloud computing credits
- **2026.04** | **Grant-in-Aid for Early-Career Scientists (KAKENHI)** —Development of an Input Data Differentiable Integrated Framework (ID3) for Biomolecular Sequence Design · ¥4,550,000; Project info
- **2025.10** | **Google research support** —Biological sequence optimization · USD 30,000
- **2024.4** | **Grant-in-Aid for Early-Career Scientists (KAKENHI)** —Development of a Large-Scale Language Model Integrating RNA Sequences and Text · ¥4,420,000; Project info

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## Publications

### Books

- **2026.08** | Hongmin Li. "*Academic Collapse Does Not Happen All at Once: A Conditional Theory of Epistemic Cascades, Expertise Cliffs, and Institutional Recoupling*" . Book · Independently published · 310 pp. · ISBN 979-8192323182. Amazon

### Preprints

- **2026.09** | Hongmin Li, Wanli Zhao. "*Where Scientific Search Agents Fail: Decision-Checkpoint Auditing of Exposure and Inspection Attempts*" . arXiv preprint · 2609.38670. arXiv
- **2026.09** | Hongmin Li. "*The Hodge Conjecture for Odd-Degree Fermat Fourfolds*" . ResearchGate preprint. ResearchGate
- **2026.09** | Hongmin Li. "*Low-t Coefficients of the Valley Delta Conjecture*" . ResearchGate preprint. ResearchGate
- **2026.09** | Linghu Fan, Hongmin Li. "*Resolutions of Linear p-Cyclic Quotient Singularities*" . arXiv preprint · 2609.07182 [math.AG]. arXiv
- **2026.06** | Hongmin Li. "*The Calibration Turn in AI-Assisted Research: A Conceptual and Methodological Framework for Evidence-Licensed Claims*" . arXiv preprint · 2606.31273 [cs.LG]. arXiv, Code & artifacts
- **2026.05** | Hongmin Li. "*Targeted Tests for LLM Reasoning: An Audit-Constrained Protocol*" . arXiv preprint · 2605.11599 [cs.LG]. arXiv
- **2026.05** | Hongmin Li. "*Separating Shortcut Transition from Cross-Family OOD Failure in a Minimal Model*" . arXiv preprint · 2605.12945 [cs.LG]. arXiv
- **2026.05** | Hongmin Li. "*A Controlled Counterexample to Strong Proxy-Based Explanations of OOD Performance: in a Fixed Pretraining-and-Probing Setup*" . arXiv preprint · 2605.11554 [cs.LG]. arXiv
- **2026.05** | Hongmin Li. "*FastUMAP: Scalable Dimensionality Reduction via Bipartite Landmark Sampling*" . arXiv preprint · 2605.11428 [cs.LG]. arXiv
- **2025.12** | Bian Bian, Yiming Zhang, Hongmin Li, Jiuzhou Zhong, Yutaka Saito. "*Large generative mRNA language foundation model for efficient coding sequence generation and design with mRNA-GPT*" . bioRxiv. Preprint
- **2023.9** | Hao Wu, Xiangjun Pei, Takashi Oguchi, Hongmin Li, Zifei Wang, Zhihao He, Runqiu Huang. "*An Integrated Physical Approach to Earthquake-Induced Landslide Susceptibility Incorporating Geological Structure: A Case Study of the Diexi Catchment, Sichuan, China*" . SSRN preprint. SSRN

### Journal Articles

- **2026.09** | Hongmin Li, Goro Terai, Takumi Otagaki, Kiyoshi Asai. "*Gradient-based Optimization for mRNA Sequence Design*" . Bioinformatics. Journal / DOI, Preprint, Code
- **2023.1** | Hongmin Li, Xiucai Ye, Akira Imakura, Tetsuya Sakurai. "*LSEC: Large-scale spectral ensemble clustering*" . Intelligent Data Analysis. arXiv, Code
- **2022.8** | Hongmin Li, Xiucai Ye, Akira Imakura, Tetsuya Sakurai. "*Divide-and-conquer based Large-Scale Spectral Clustering*" . Neurocomputing. Preprint, Code
- **2022.7** | Meng Huang, Xiucai Ye, Hongmin Li, Tetsuya Sakurai. "*Missing Value Imputation With Low-Rank Matrix Completion in Single-Cell RNA-Seq Data by Considering Cell Heterogeneity*" . Frontiers in Genetics. DOI
- **2020.7** | Xiucai Ye, Hongmin Li, Akira Imakura, Tetsuya Sakurai. "*An Oversampling Framework for Imbalanced Classification Based on Laplacian Eigenmaps*" . Neurocomputing. Link
- **2019** | Xiucai Ye, Hongmin Li, Tetsuya Sakurai, Pei-Wei Shueng. "*Ensemble Feature Learning to Identify Risk Factors for Predicting Secondary Cancer*" . International Journal of Medical Sciences. DOI

**Conference Papers**

- **2020.11** | Hongmin Li, Xiucai Ye, Akira Imakura, Tetsuya Sakurai. "*Ensemble Learning for Spectral Clustering*" . ICDM 2020. PDF, Code
- **2020.7** | Hongmin Li, Xiucai Ye, Akira Imakura, Tetsuya Sakurai. "*Hubness-based Sampling Method for Nyström Spectral Clustering*" . IJCNN 2020. Link
- **2019.8** | Xiucai Ye, Hongmin Li, Akira Imakura, Tetsuya Sakurai. "*Distributed Collaborative Feature Selection Based on Intermediate Representation*" . IJCAI 2019. Link
- **2018.10** | Xiucai Ye, Hongmin Li, Tetsuya Sakurai, Zhi Liu. "*Large Scale Spectral Clustering Using Sparse Representation Based on Hubness*" . CBDCom 2018. Link, Code

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## AI-generated mathematics (rStation)

AI-generated mathematical manuscripts released for human verification on rStation, an online platform for research preprints with platform review. Accepted on rStation means accepted after rStation platform review; it is not journal peer review. The manuscripts were generated with AI assistance and their proofs require independent human verification. None of these papers claims to resolve the Riemann Hypothesis, the Hodge conjecture or P versus NP.

- **2026.10.27** | "*Exponentially Many Barnette Graphs with the Global $H^{+-}$ Property*" . rStation r/math, math_1 v2 (accepted after platform review). rStation
- **2026.10.26** | "*A Positive Reciprocal-Product Criterion for the Riemann Hypothesis*" . rStation r/riemann_hypothesis, riemann_hypothesis_42 v2 (accepted after platform review). rStation
- **2026.10.26** | "*Generalized Thomas Sections and Density-Corrected Residue Selectors*" . rStation r/hodge_conjecture, hodge_conjecture_39 v2 (accepted after platform review). rStation
- **2026.10.26** | "*A Single Modular Character Is Complete for SYM∘LTF Counting*". rStation r/computer_science, computer_science_1 v2 (accepted after platform review). rStation
- **2026.10.25** | "*Truth-Table Quotients Do Not Close One-Tape MCSP Magnification: Small Fibers, Exact Cosets, and a Defect-Inflation Dichotomy*" . rStation r/p_vs_np, p_vs_np_1 v1 (accepted after platform review). rStation
- **2026.10.25** | "*Tight Ordered Width for Exponential-Threshold MCSP*" . rStation r/computer_science, computer_science_2 v1 (accepted after platform review). rStation
- **2026.10.25** | "*Concentration of Random Tensor Sumsets*" . rStation r/math, math_2 v1 (accepted after platform review). rStation
- **2026.10.23** | "*Cutoff-free positivity of localized Weil forms through the (log 11)/2 threshold*" . rStation r/riemann_hypothesis, riemann_hypothesis_45 v1 (accepted after platform review). rStation
- **2026.10.22** | "*Balanced Cyclic Prym Families and the Generalized Hodge Conjecture for All Powers*". rStation r/hodge_conjecture, hodge_conjecture_38 v1 (accepted after platform review). rStation
- **2026.10.22** | "*Moving Apolar Constraints on Hodge Cycles of the Fermat Sextic Fourfold*" . rStation r/hodge_conjecture, hodge_conjecture_41 v1 (accepted after platform review). rStation
- **2026.10.19** | "*A small-diagonal transfer from a degree-33 Fermat character to the quartic-CM determinant of a Prym eightfold, and the Hodge conjecture for its powers under explicit quotient hypotheses*" . rStation r/hodge_conjecture, hodge_conjecture_20 v3 (accepted after platform review). Conditional on explicit quotient hypotheses stated in the paper. rStation

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## Conference presentations

- **2026.06.09** | **Poster presentation —JST CREST BioDX Area Meeting** —5th area meeting · Numazu, Japan
- **2025.11** | **Poster presentation —JST CREST BioDX Interim Symposium** —Event
- **2025.8** | **Oral presentation —Input Data Differentiable Designer (ID3) at RNA Informatics Dojo 2025** — Early presentation of the method later developed into the preprint "Gradient-based Optimization for mRNA Sequence Design" ; Event, Paper, Code
- **2024.10** | **Poster presentation —FastUMAP at the Asia-Pacific Bioinformatics Joint Conference 2024** —Early conference presentation of the work later developed into the FastUMAP preprint; Event, Paper
- **2024.8** | **Oral presentation —RNA Informatics Dojo 2024**
- **2024.8** | **Participation —IJCAI 24 (Jeju)**
- **2024.6** | **Participation —WCCI 24 (Yokohama)**
- **2024.6** | **Participation —25th Annual Meeting of the RNA Society of Japan**
- **2023.8** | **Oral presentation —RNA Informatics Dojo 2023**
- **2022.8** | **Oral presentation —RNA Informatics Dojo 2022**
- **2022.3** | **Presentation —JST SPRING recipients event**
- **2021.12** | **Oral presentation —CollaboTICS 2021**
- **2019.6** | **Poster presentation —G20 Digital Science Now Symposium**

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## Peer Review Activities

- **2026.07** | **Conference on Neural Information Processing Systems (NeurIPS)**
- **2026.07** | **Neurocomputing**
- **2026.05** | **Journal of King Saud University - Computer and Information Sciences**
- **2026.01** | **The Journal of Supercomputing**
- **2025** | **International Journal of Machine Learning and Cybernetics**
- **2024.11** | **Pattern Analysis and Applications**
- **2024.11** | **Knowledge and Information Systems**
- **2024.10** | **Neurocomputing**
- **2024.09** | **The Visual Computer**
- **2024.03** | **International Journal of Machine Learning and Cybernetics**
- **2022.11** | **Transactions on Pattern Analysis and Machine Intelligence**
- **2022.10** | **Briefings in Functional Genomics**
- **2021–2022** | **Pattern Recognition**

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## Honors & fellowships

- **2021.10** | **Doctoral research support —JST SPRING** —Support for Pioneering Research Initiated by the Next Generation
- **2020.10** | **AETA Earthquake Prediction AI Algorithm Competition 2019 —Second Prize** —Second-place recognition for an AI model predicting earthquakes from real-time monitoring data · awarded in 2020; News, Slides
- **2019.10** | **3rd Analysys International Algorithm Competition —Special Award** —Special recognition in the website page-view and unique-visitor prediction task
- **2018.10** | **IEEE International Conference on Cloud and Big Data Computing 2018 —Best Paper Award** —Awarded for "Large Scale Spectral Clustering Using Sparse Representation Based on Hubness"

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## Skills

### Programming Languages

- Python, R, MATLAB, C/C++, Java

**Machine Learning & Deep Learning**

- PyTorch, TensorFlow, scikit-learn
- Spectral Clustering, Large Language Models
- Bioinformatics Applications

**Computational Biology**

- RNA Sequence Analysis
- mRNA Design and Optimization
- Single-cell RNA-seq Data Analysis